THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

Deconstructing where time is lost: Nominal P50 (52.2 ms) vs. P99 Latency Tail Blowout (126.0 ms) Breaching World Deadlines

Systems Insight: While mean latency (*P50 = 52.2 ms*) sits comfortably within stability margins, real-world memory crossbar contention and OS thread preemption inflate *P99* to 126.0 ms, breaching the physical deadline and forcing open-loop drift unless arrested by a 1 kHz MCU safety reflex.